

Instructions for *ACL Proceedings

Anonymous ACL submission

1 Task Description

The task in this project is to perform a membership inference attack (MIA) on finetuned language models. More specifically, we aim to predict whether a given sample was included in the models' finetuning dataset by assigning each sample a membership score. This task is important because it helps evaluate whether a model leaks information about its training data, which is closely related to privacy and memorization risks in machine learning systems. Therefore, the goal of this project is to develop a method that can distinguish member samples from non-member samples as accurately as possible and submit these predictions to the Kaggle competition.

2 Background

Membership Inference Attacks (MIAs), as an approach to identify whether a specific data point belongs to a target model's training dataset, have been widely utilized, primarily in traditional machine learning models (?). Although some studies (??) indicate that MIAs are effective on LLMs, recent research disproves this argument, demonstrating that MIAs barely outperform random guessing when evaluating the pre-training data of large language models (?).

? conducted a performance analysis of MIAs on various models ranging from 70 million to 12 billion parameters. The evaluation targeted the following models:

- Pythia: 70M, 160M, 1.4B, 2.8B, 6.9B, 12B
- GPT-Neo: 125M, 1.3B, 2.7B
- Datablations: 2.8B
- SILO: 1.3B
- OLMo: 1B, 7B

The evaluation was performed on training sets composed of Wikipedia, PubMed Central, ArXiv, Pile-CC, Github, DM Math, and HackerNews. The research concludes that MIA's unreliable performance in evaluating data leakage is primarily due to two factors: first, the large scale of training data combined with near-one-epoch training limits model overfitting; second, there is an inherently fuzzy boundary, or high n-gram overlap, between member data and non-member data.

Furthermore, ? proposed an alternative methodology to bypass the reliance on reference datasets by conducting a Neighbourhood Attack. This attack was evaluated during the fine-tuning phase on the 117M parameter version of GPT-2. The target model was fine-tuned on 60,000 samples from the AG News corpus, 150,000 samples from Twitter, and 100,000 samples from Wikitext-103, while the neighbour texts were dynamically generated using a pre-trained BERT model. The paper finds that the Neighbourhood attack can outperform traditional reference-based MIAs (e.g., LiRA) by up to 100% under realistic assumptions.

Moreover, ? advance this line of research by introducing SPV-MIA (Self-calibrated Probabilistic Variation MIA), an architecture composed of two main modules: a Self-prompt Reference Model and a Probabilistic Variation Assessment[cite: 19, 80]. The authors evaluated this method on four target models (GPT-2, GPT-J, Falcon-7B, and LLaMA-7B) fine-tuned on the Wikitext-103, AG News, and XSum datasets[cite: 268]. The results demonstrate that SPV-MIA is highly robust and effectively bypasses the unrealistic reference data assumptions of prior reference-based attacks[cite: 83, 92]. Specifically, it significantly elevates attack performance, increasing the AUC from approximately 0.7 to an average of 0.924 [cite: 23, 287], while achieving an average True Positive Rate (TPR) of 46.9% at a 1% False Positive Rate (FPR).

3 Reflection

Darwin Zhang:

- What do you expect to gain from this shared task and what did you actually gain?

From this task, I was expecting a challenge and to see where our teamwork skills will lead us. By challenge, I mean as we do not have set methods on what we are performing, but allow us to personally interpret and find papers to communicate and attempt to solve the problem. From the actual gains, I believe I learned how to communicate better to my group and work with them. From spreading out work, to having meetings on how to approach each week for the project, I learned a lot from my group mates and how to work with them. In terms from the lab content, I learned that many methods are somewhat similar, but performance changes a lot. Therefore, it is great to do many tests and to understand the methods more in depth to try and understand why it works and what makes it different from others.

- Does your task go well? What goes well and what doesn't go well? What is the biggest barrier? Do you feel that you are on the right track?

I believe this task went well overall from how we structured and approached it. Structuring, organization, and communications were the strong suits towards our project and allowed us to get a relatively great performance at the end. What did not go as well is that we did most of the work closer to the deadline, but this is expected as we have many other things due. This is not to say that we were all last minute, but more so towards the idea that this project did take more time than we expected and having more buffer time in case something did not work would be more optimal for future iterations. So I would say that we are on the right track and performing well, but now since our performance is on the higher end from our initial attempts and the baseline, the real barrier now is how do we improve it. This can be from finding more papers and experimenting with methods, and this can more or less be attributed to having time as our barrier.

- If you could re-start this task again, what are the good practices you want to keep and what are the not-so-good practices you want to improve?

For future iterations, one really good practice I would keep is our communication and structuring. Having this foundation, it makes it really easy to contact each other if needed, ask for help, or let us know if there is a problem. The structure has really made dividing up tasks easy as well and we all had an attempt towards each part of the project from reading papers to training. For no-so-good practices, it was not a huge problem as stated before, but I believe we should look into time management and starting earlier, even a little bit, so we are able to improve performance over the week, instead of towards the deadline where we are not able to train and submit a lot for one day.

Yusen Huang:

- What do you expect to gain from this shared task and what did you actually gain? My expectation for this project was to learn some methods that can help me with the membership inference attacks, and I think from my study and what my teammates' study I did learn a great deal of what kind of methods we should use to tackle this problem. I used the "Min-K% Prob" method to help with the task and by implementing it I gained firsthand experience on how to complete such a goal.

- Does your task go well? What goes well and what doesn't go well? What is the biggest barrier? Do you feel that you are on the right track? Admittedly, before Milestone 2, the methods we used were not good enough. All our AUC scores fell short compared to other groups' results and we were left behind. However, I think this is also because we were trying to approach this project one step at a time instead of just using the best method to deal with the problem. Our tactic was using relatively simpler methods that require a low level of training, and while they did improve compared to the baseline model, the results were not good enough. We also tried some more complicated methods during Milestone 2 but

173	were not able to complete them due to limited		
174	computational resources. text		
175			
176	• If you could re-start this task again, what are		
177	the good practices you want to keep and what		
178	are the not-so-good practices you want to im-		
179	prove? I think the improvements that can be		
180	made for this project are that we need to focus		
181	more on paper research and team synchroniza-		
182	tion. We were essentially working indepen-		
183	dently without enough cross-communication		
184	about our findings and results. If we had com-		
185	municated more, I think we probably would		
186	have had better scores. That said, I think we		
187	also did have some good practices in how we		
188	set up the project and how we handled the dis-		
189	tribution of work. Every team member knew		
190	exactly what they were doing during the mile-		
191	stone and there was no overlapping in work		
192	and methods. text		
193			
194	Marco Wang:		
195	• What do you expect to gain from this shared		
196	task and what did you actually gain?		
197	I expected to learn how Membership Inference		
198	Attacks (MIA) are applied to Large Language		
199	Models in practice, especially how to turn the		
200	theory into working code. I also wanted to		
201	better understand where LLMs are vulnerable.		
202	In the end, I gained hands-on experience im-		
203	plementing the Base Model Likelihood Ratio		
204	(Reference Model) approach. This helped me		
205	understand how to separate true memorization		
206	from the general linguistic difficulty of clini-		
207	cal text by comparing the fine-tuned model's		
208	loss with a base model's loss. I also gained		
209	experience in coordinating technical work and		
210	collaborating within a team.		
211			
212	• Does your task go well? What goes well and		
213	what doesn't go well? What is the biggest		
214	barrier? Do you feel that you are on the right		
215	track?		
216	Our task did not go well at first. In Milestone		
217	1 and early Milestone 2, our baseline methods		
218	based only on sequence-level loss performed		
219	poorly because they were strongly affected		
220	by the complexity of the medical texts. As		
221	a result, we fell behind. Our biggest barriers		
	were limited computational resources and		
	time, which made it difficult to test more ad-		
	vanced ideas. In Milestone 2, we focused on		
	improving our best model with methods that		
	did not require heavy training or major re-		
	design, but the results were still weaker than		
	those of other students. However, in Mile-		
	stone 3, we managed to address these issues		
	and significantly improve our performance.		
	Because of this progress, I believe we are now		
	on the right track, even though the process		
	was challenging.		
	• If you could re-start this task again, what are		
	the good practices you want to keep and what		
	are the not-so-good practices you want to im-		
	prove?		
	If I could restart this task, I would keep our		
	clear division of labor. Giving each mem-		
	ber a separate focus such as trying out dif-		
	ferent models/methods, or literature review.		
	This helped us explore multiple directions ef-		
	ficiently. However, I would improve our team		
	coordination and timeline management. We		
	worked too independently and left too much		
	experimentation until late in the project. Next		
	time, I would set up earlier and more frequent		
	check-ins so we could share findings sooner		
	and improve our methods with less time pres-		
	sure.		
	Tianhao Cao:		
	• What do you expect to gain from this shared		
	task and what did you actually gain?		
	My primary expectation for this shared task		
	was to acquire practical, real-world experience		
	with membership inference attacks (MIAs).		
	Through the course of this project, I success-		
	fully achieved this goal. Immersing myself		
	in the task allowed me to bridge the gap be-		
	tween theoretical knowledge and practical ap-		
	plication, giving me a comprehensive under-		
	standing of MIA methodologies and their real-		
	world implications.		
	• Does your task go well? What goes well and		
	what doesn't go well? What is the biggest		
	barrier? Do you feel that you are on the right		
	track?		
	Overall, the project progressed smoothly as		
	we systematically met our established mile-		
	stones. However, the initial phases presented		
	significant challenges. I encountered consider-		
	able difficulty implementing specific method-		
	ologies, including the Neighborhood Attack,		

Casing Attack, and Metric/Threshold-based attacks; none of my initial models yielded satisfactory AUC or TPR@1%FPR metrics. The biggest barrier was figuring out how to optimize these baseline models. After conducting a deeper review of relevant literature, I successfully overcame this hurdle by incorporating a 360M parameter model into the comparison for the Neighborhood Attack. This adjustment significantly improved the results and reassured me that we were indeed on the right track.

- If you could re-start this task again, what are the good practices you want to keep and what are the not-so-good practices you want to improve?

If I were to restart this task, a key practice I would definitely retain is conducting extensive literature reviews. Reading as many relevant papers as possible was instrumental in building a strong foundational understanding. It provided crucial insights into which approaches had historically failed and which had succeeded, allowing me to build upon existing research rather than reinventing the wheel.

Conversely, a critical area for improvement is my code testing protocol. Early in the project, I made the mistake of executing computationally heavy algorithms without running thorough preliminary tests on smaller subsets of data. My first major script ran for two hours only to return incorrect results, resulting in a significant waste of time. Moving forward, I will strictly enforce modular testing and debug on small sample sizes before committing to full-scale executions.